



Machine Learning

AIML CZG565

M8 : Bayesian Learning

BITS Pilani

Pilani Campus

Course Faculty of M.Tech Cluster
BITS – CSIS - WILP

Disclaimer and Acknowledgement



- These content of modules & context under topics are planned by the course owner Dr. Sugata, with grateful acknowledgement to many others who made their course materials freely available online
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas
- We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation

Slide Source / Preparation / Review:

From BITS Pilani WILP: Prof.Sugata, Prof.Chetana, , Prof.Monali, Prof.Rajavadhana

External: CS109 and CS229 Stanford lecture notes, Dr.Andrew NG and many others who made their course materials freely available online

Course Plan

M1	Introduction & Mathematical Preliminaries
M2	Machine Learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning
M7	Support Vector Machine – Planned to be discussed in CS 15 or CS 16
M8	Bayesian Learning
M9	Ensemble Learning ✓
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

Text Classification using Naive Bayes Classifier

The Bag of Words Representation



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsic	1
al	
times	1
sweet	1
satirical	1
adventur	1
e	
genre	1
fairy	1
humor	1
have	1
great	1
...	..

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



C1
C2
C3

w1 w2 w3

Example : Multinomial model :

Which Tag sentence "A very close game" belong to?

- ✓ $P(\text{Sports}=\text{Yes}) = 3/5$
- ✓ $P(\text{Sports}=\text{No}) = 2/5$
- $P(A \mid \text{Sports}=\text{Yes}) = 2/11$
- $P(A \mid \text{Sports}=\text{No}) = 1/9$

$$\begin{aligned}
 P(\text{Sports}=\text{Yes} \mid X) &= P(X \mid \text{Sports}=\text{Yes}) \cdot P(\text{Sports}=\text{Yes}) / P(X) \\
 &= P(X \mid \text{Sports}=\text{Yes}) \cdot (3/5) \\
 &= P(A \mid \text{Sports}=\text{Yes}) \cdot P(\text{Very} \mid \text{Sports}=\text{Yes}) \cdot P(\text{Close} \mid \text{Sports}=\text{Yes}) \cdot P(\text{Game} \mid \text{Sports}=\text{Yes}) \cdot (3/5) \\
 &= (2/11) \cdot (1/11) \cdot (0/11) \cdot (2/11) \cdot (3/5)
 \end{aligned}$$

Text	Tag
"A great game"	Sports ✓
"The election was over"	Not sports
"Very clean match"	Sports ✓
"A clean but forgettable game"	Sports ✓
"It was a close election"	Not sports

A	Great	Game	The	Election	Was	Over	Very	Clean	Match	But	Forgettable	It	Close	Sports or Not Sports
1 ✓	1	1												1 ✓
			1	1	1	1								0
							1	1	1					1
1 ✓		1						1		1	1			1
1				1	1							1	1	0
1		1					1						1	????

$$\underline{P(\text{Yes} | X)} = \frac{P(X | \text{Yes}) P(\text{Yes})}{P(X)}$$

$$P(\text{Yes}) = 3/5$$

$$= P(A/Y) \cdot P(\text{very}/Y) \cdot P(\text{close}/Y) \cdot P(\text{game}/Y) \times P(\text{Yes}).$$

$\frac{\# A(\text{Yes})}{\text{Tot. \# features in Yes.}}$

$$P(\text{No} | X) = \frac{P(X | \text{No}) P(\text{No})}{P(X)}$$

$$P(\text{No}) = \frac{2}{5}$$

$X = A \quad \text{very} \quad \text{close} \quad \text{game}$

Laplace Smoothing

$$\frac{2+1}{11+14}$$

- Laplace smoothing: we add 1 or in general constant k to every count so it's never zero.
- To balance this, we add the number of possible words to the divisor, so the division will never be greater than 1
- In our case, the possible words are ['a', 'great', 'very', 'over', 'it', 'but', 'game', 'election', 'clean', 'close', 'the', 'was', 'forgettable', 'match'].
- In our example
- we add 1 to every probability, therefore the probability, such as **P(close | sports)**, will never be 0

$$\frac{N_{ic} + \alpha}{N_c + \alpha d}$$

$\alpha = 2$ $\frac{2+2}{11+2(14)}$ $\alpha = 1$ $d = \text{Vocabulary size}$ $\rightarrow \text{distinct}$

Example : Multinomial model :

Which Tag sentence “ A very close game” belong to?

$$\begin{aligned}
 P(\text{Sports}=\text{Yes}) &= 3/5 \\
 P(\text{Sports}=\text{No}) &= 2/5 \\
 P(A \mid \text{Sports}=\text{Yes}) &= 2/11 \\
 P(A \mid \text{Sports}=\text{No}) &= 1/9
 \end{aligned}$$

$$\alpha = 1 \qquad \frac{0+1}{11+14}$$

$$\begin{aligned}
 P(\text{Sports}=\text{Yes} \mid X) &= P(X \mid \text{Sports}=\text{Yes}). P(\text{Sports}=\text{Yes}) / P(X) \\
 &= P(X \mid \text{Sports}=\text{Yes}). (3/5) \\
 &= P(A \mid \text{Sports}=\text{Yes}). P(\text{Very} \mid \text{Sports}=\text{Yes}). P(\text{Close} \mid \text{Sports}=\text{Yes}). P(\text{Game} \mid \text{Sports}=\text{Yes}). (3/5) \\
 &= (2/11) \cdot (1/11) \cdot (0/11) \cdot (2/11) \cdot (3/5) \\
 &= (2+1/11+14) \cdot (1+1/11+14) \cdot (0+1/11+14) \cdot (2+1/11+14) \cdot (3/5) \\
 &= 0.00002765
 \end{aligned}$$

(3+14/5+28)

=0.00002396

A	Great	Game	The	Election	Was	Over	Very	Clean	Match	But	Forgettable	It	Close	Sports or Not Sports
1	1	1												1
			1	1	1	1								0
							1	1	1					1
1		1						1		1	1			1
1				1	1							1	1	0
1		1					1						1	????

Apply Laplace Smoothing

Word	P(word Sports)	P(word Not Sports)
a	2+1 / 11+14	1+1 / 9+14
very	1+1 / 11+14	0+1 / 9+14
close	0+1 / 11+14	1+1 / 9+14
game	2+1 / 11+14	0+1 / 9+14

$$\begin{aligned}
 &P(a|Sports) \times P(very|Sports) \times P(close|Sports) \times P(game|Sports) \times \\
 &P(Sports) \\
 &= 2.76 \times 10^{-5} \\
 &= 0.0000276 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 &P(a|Not Sports) \times P(very|Not Sports) \times P(close|Not Sports) \times \\
 &P(game|Not Sports) \times P(Not Sports) \\
 &= 0.572 \times 10^{-5} \\
 &= 0.00000572 \quad \checkmark
 \end{aligned}$$

Example 2: Multinomial model

	docID	words in document	in $c = \text{China?}$
training set	1	Chinese Beijing Chinese	yes
	2	Chinese Chinese Shanghai	yes
	3	Chinese Macao	yes
	4	Tokyo Japan Chinese	no
test set	5	Chinese Chinese Chinese Tokyo Japan	?

$$\hat{P}(w_i | C_k) = \frac{n_k(w_i)}{\sum_{j=1}^{|V|} n_k(w_j)}$$

Chinese
Yes $\rightarrow 5$
No $\rightarrow 1$

$d = 6$ ✓

$P(\text{Yes})$

$P(\text{No})$

$T\#Y = 8$

$T\#N = 3$

$N_{\text{yes}}(W=\text{Chinese}) = 5$, $N_{\text{No}}(W=\text{Chinese}) = 1$,

$|V| = 6 = \{\text{Chinese, Beijing, Shanghai, Macao, Tokyo Japan}\}$

No of features (words) in Yes class = 8

No of features (words) in No class = 3

Source Credit : <https://nlp.stanford.edu/IR-book>

$$P(\text{Yes} | X) = \overset{CCTJ}{P(X | Y_{\sim})} P(Y_{\sim}) \quad 3/4$$

$$P(C/Y) P(C/Y) P(C/Y) P(T/Y) P(J/Y) P(Y)$$

$$P(No | X) = \frac{5+1}{8+6} \frac{5+1}{8+6} \frac{5+1}{8+6} \frac{0+1}{8+6} \frac{0+1}{8+6} \times \frac{3}{4}$$

Example 2

	docID	words in document	in $c = \text{China}$?
training set	1	Chinese Beijing Chinese	yes
	2	Chinese Chinese Shanghai	yes
	3	Chinese Macao	yes
	4	Tokyo Japan Chinese	no
test set	5	Chinese Chinese Chinese Tokyo Japan	?

Priors: $\hat{P}(c) = 3/4$ and $\hat{P}(\bar{c}) = 1/4$

$$\hat{P}(w_t | C_k) = \frac{n_k(w_t)}{\sum_{s=1}^{|V|} n_k(w_s)},$$

$$\hat{P}(\text{CHINESE}|c) = (5 + 1)/(8 + 6) = 6/14 = 3/7$$

$$\hat{P}(\text{TOKYO}|c) = \hat{P}(\text{JAPAN}|c) = (0 + 1)/(8 + 6) = 1/14$$

$$\hat{P}(\text{CHINESE}|\bar{c}) = (1 + 1)/(3 + 6) = 2/9$$

$$\hat{P}(\text{TOKYO}|\bar{c}) = \hat{P}(\text{JAPAN}|\bar{c}) = (1 + 1)/(3 + 6) = 2/9$$

Source Credit : <https://nlp.stanford.edu/IR-book>

Example 2

	docID	words in document	in $c = \text{China}$?
training set	1	Chinese Beijing Chinese	yes
	2	Chinese Chinese Shanghai	yes
	3	Chinese Macao	yes
	4	Tokyo Japan Chinese	no
test set	5	Chinese Chinese Chinese Tokyo Japan	?

$$P(C_k | \mathcal{D}) \propto P(C_k) \prod_{j=1}^{\text{len}(\mathcal{D})} P(u_j | C_k) \quad \text{u-each word in test document}$$

$$\hat{P}(c | d_5) \propto 3/4 \cdot (3/7)^3 \cdot 1/14 \cdot 1/14 \approx 0.0003 \quad \checkmark$$

$$\hat{P}(\bar{c} | d_5) \propto 1/4 \cdot (2/9)^3 \cdot 2/9 \cdot 2/9 \approx 0.0001 \quad \checkmark$$

Naïve Bayes classifier: Summary Model



Model: joint probability distribution given by

- $P(X, Y) = P(Y) P(X|Y)$
- $P(X = X_1, \dots, X_n, Y = y_k) = P(Y = y_k) P(X = X_1, \dots, X_n | Y = y_k)$

[mixed-naïve-
-bayes]

Learning/Training:

For output variable Y

- $P(Y) \sim \text{Ber}(p)$

Binary 0/1

For each attribute X

- $P(X|Y) \sim \text{Ber}(\pi) \rightarrow$ Multivariate Bernoulli NB (X_i – binary)
- $P(X|Y) \sim \text{Multinom}(\pi, n) \rightarrow$ Multinomial NB (X_i – multinomial)
- $P(X|Y) \sim N(\mu, \sigma^2) \rightarrow$ GaussianNB (X_i – real valued)

categorical NB

Source Credit : <https://nlp.stanford.edu/IR-book>

Logistic Regression vs Naïve Bayes

MLE

MAP



Idea:

- Naïve Bayes allows computing $P(Y|X)$ by learning $P(Y)$ and $P(X|Y)$
- Why not learn $P(Y|X)$ directly?

Logistic Regression and Gaussian Naïve Bayes Classifier



MLE

- Interestingly, the parametric form of $P(Y|X)$ used by Logistic Regression is precisely the form implied by the assumptions of a Gaussian Naive Bayes classifier.
- Therefore, we can view Logistic Regression as a closely related alternative to GNB, though the two can produce different results in many cases
- Derivation given in CS-5 slides in self learning section and also attached PDF of new chapter from Tom Mitchell book

Features of Bayesian learning

- Each observed training example can **incrementally decrease or increase the estimated probability** that a hypothesis is correct.
- Flexible approach to learning than algorithms that **completely eliminate a hypothesis** if it is found to be inconsistent with any single example.

Practical Issues of Bayesian learning

- Require initial knowledge of many probabilities
 - Often estimated based on background knowledge, previously available data, and assumptions about the form of the underlying distributions.
- Significant computational cost required to determine the Bayes optimal hypothesis in the general case (linear in the number of candidate hypotheses)

References

- <https://www.inf.ed.ac.uk/teaching/courses/inf2b/learnnotes/inf2b-learn07-notes-nup.pdf>
- <https://cs229.stanford.edu/summer2019/cs229-notes2.pdf>
- Tom Mitchell – Chapter 6
- <https://nlp.stanford.edu/IR-book/pdf/13bayes.pdf>

Logistic Regression & Naïve Bayes

Self Read

(Proof : Mathematical Relation between Logistic Regression and Naïve Bayes)

Where does the **form** come from?

- Logistic regression hypothesis representation

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}} = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n)}}$$

- Consider learning $f: X \rightarrow Y$, where
 - X is a vector of real-valued features $[X_1, \dots, X_n]^T$
 - Y is Boolean
 - Assume all X_i are conditionally independent given Y
 - Model $P(X_i|Y = y_k)$ as Gaussian $N(\mu_{ik}, \sigma_i)$
 - Model $P(Y)$ as Bernoulli π

What is $P(Y|X_1, X_2, \dots, X_n)$?

Slide credit: Tom Mitchell

Where does the **form** come from?

- $$P(Y = 1|X) = \frac{P(Y=1)P(X|Y=1)}{P(Y=1)P(X|Y=1) + P(Y=0)P(X|Y=0)}$$

Applying Bayes rule

$$= \frac{1}{1 + \frac{P(Y=0)P(X|Y=0)}{P(Y=1)P(X|Y=1)}}$$

Divide by $P(Y = 1)P(X|Y = 1)$

$$= \frac{1}{1 + \exp(\ln(\frac{P(Y=0)P(X|Y=0)}{P(Y=1)P(X|Y=1)}))}$$

Apply $\exp(\ln(\cdot))$

$$= \frac{1}{1 + \exp(\ln(\frac{1-\pi}{\pi}) - \sum_i \ln \frac{P(X_i|Y=0)}{P(X_i|Y=1)})}$$

Plug in $P(X_i|Y)$

$$P(x|y_k) = \frac{1}{\sqrt{2\pi}\sigma_i} e^{-\frac{(x-\mu_{ik})^2}{2\sigma_i^2}}$$

$$\sum_i \left(\frac{\mu_{i0} - \mu_{i1}}{\sigma_i^2} X_i + \frac{\mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2} \right)$$

$$P(Y = 1|X_1, X_2, \dots, X_n) = \frac{1}{1 + \exp(\theta_0 + \sum_i \theta_i X_i)}$$

Slide credit: Tom Mitchell

Where does the **hypothesis function** come from?

- Logistic regression hypothesis representation

$$P(Y=1|X) = h_{\theta}(x) = \frac{1}{1+e^{-\theta^T x}} = \frac{1}{1+e^{-(\theta_0+\theta_1x_1+\theta_2x_2+\dots+\theta_nx_n)}}$$

- Model likelihood $P(X_i|Y = y_k)$ as Gaussian $N(\mu_{ik}, \sigma_i)$ and assume variance is independent of class, i.e. $\sigma_{i0} = \sigma_{i1} = \sigma_i$

$$P(x|y_k) = \frac{1}{\sqrt{2\pi}\sigma_i} e^{-\frac{(x-\mu_{ik})^2}{2\sigma_i^2}}$$

- Model prior $P(Y)$ as Bernoulli π : $P(Y=1) = \pi$ and $P(Y=0) = 1-\pi$

What is $P(Y|X_1, X_2, \dots, X_n)$?

Logistic Regression –Bayesian Analysis

$$P(Y = 1|X) = \frac{P(Y = 1)P(X|Y = 1)}{P(Y = 1)P(X|Y = 1) + P(Y = 0)P(X|Y = 0)}$$

Applying Bayes rule

$$P(Y = 1|X) = \frac{1}{1 + \frac{P(Y=0)P(X|Y=0)}{P(Y=1)P(X|Y=1)}}$$

Divide by $P(Y = 1)P(X|Y = 1)$

$$P(Y = 1|X) = \frac{1}{1 + \exp(\ln \frac{P(Y=0)P(X|Y=0)}{P(Y=1)P(X|Y=1)})}$$

Apply $\exp(\ln(\cdot))$

$$P(Y = 1|X) = \frac{1}{1 + \exp \left(\ln \frac{P(Y = 0)}{P(Y = 1)} + \ln \frac{P(X|Y = 0)}{P(X|Y = 1)} \right)}$$

Logistic Regression –Bayesian Analysis

By independence assumption:

$$\frac{P(X|Y = 0)}{P(X|Y = 1)} = \prod_i \frac{P(X_i|Y = 0)}{P(X_i|Y = 1)}$$

$P(Y=1)=\pi$ and $P(Y=0)=1-\pi$
by modelling $P(Y)$ as Bernoulli

$$P(Y = 1|X) = \frac{1}{1 + \exp \left(\ln \frac{1-\pi}{\pi} + \ln \prod_i \frac{P(X_i|Y = 0)}{P(X_i|Y = 1)} \right)}$$

$$P(Y = 1|X) = \frac{1}{1 + \exp \left(\ln \frac{1-\pi}{\pi} + \sum_i \ln \frac{P(X_i|Y = 0)}{P(X_i|Y = 1)} \right)}$$

Logistic Regression –Bayesian Analysis



Plug in $P(X_i|Y)$

$$P(Y = 1|X) = \frac{1}{1 + \exp(\ln \frac{1-\pi}{\pi} + \sum_i \left(\frac{\mu_{i0} - \mu_{i1}}{\sigma_i^2} X_i + \frac{\mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2} \right))}$$

$$P(Y = 1|X) = \frac{1}{1 + \exp(w_0 + \sum_{i=1}^n w_i X_i)}$$

$$w_0 = \ln \frac{1-\pi}{\pi} + \sum_i \frac{\mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2}$$

$$w_i = \frac{\mu_{i0} - \mu_{i1}}{\sigma_i^2}$$

$$\begin{aligned} \sum_i \ln \frac{P(X_i|Y=0)}{P(X_i|Y=1)} &= \sum_i \ln \frac{\frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(\frac{-(X_i - \mu_{i0})^2}{2\sigma_i^2}\right)}{\frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(\frac{-(X_i - \mu_{i1})^2}{2\sigma_i^2}\right)} \\ &= \sum_i \ln \exp\left(\frac{(X_i - \mu_{i1})^2 - (X_i - \mu_{i0})^2}{2\sigma_i^2}\right) \\ &= \sum_i \left(\frac{(X_i - \mu_{i1})^2 - (X_i - \mu_{i0})^2}{2\sigma_i^2}\right) \\ &= \sum_i \left(\frac{(X_i^2 - 2X_i\mu_{i1} + \mu_{i1}^2) - (X_i^2 - 2X_i\mu_{i0} + \mu_{i0}^2)}{2\sigma_i^2}\right) \\ &= \sum_i \left(\frac{2X_i(\mu_{i0} - \mu_{i1}) + \mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2}\right) \\ &= \sum_i \left(\frac{\mu_{i0} - \mu_{i1}}{\sigma_i^2} X_i + \frac{\mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2}\right) \end{aligned}$$

Some Additional References

- T1 book by Tom Mitchell – Chapter-6
- <https://web.stanford.edu/class/archive/cs/cs109>
- https://cs229.stanford.edu/lectures-spring2022/main_notes.pdf
- <https://www.cs.cmu.edu/~ninamf/courses/601sp15/lectures.shtml>
- <https://nlp.stanford.edu/IR-book> - Chapter 13

Thank you !

Required Reading for completed session :

T1 - Chapter # 8 (Tom M. Mitchell, Machine Learning)

T1 - Chapter # 6 (Tom M. Mitchell, Machine Learning)

R1 – Chapter # 3 (Christopher M. Bhisop, Pattern Recognition & Machine Learning)

& Refresh your MFDS & ISM parallel course basics

Next Session Plan :

Module 9 : Ensemble Learning